Creating nanostructured targets to increase laser absorption and ion acceleration for inertial fusion applications

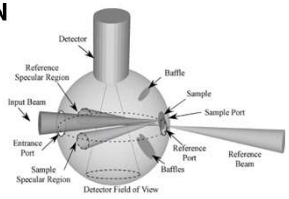
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ABSTRACT

Inertial fusion is of great interest as a sustainable energy source, with efficient laser light absorption by the target being a critical factor in optimizing inertial fusion processes. This study focuses on the fabrication and analysis of polyethylene and nickel nanowires, which enhance laser-matter interactions relevant to fusion applications. The research aims to improve nanowire fabrication process such that nanowire arrays can reliably be made more uniformly with thin substrates. Uniform and thin substrate arrays of nanowires will be crucial in experiments which seek to understand the interaction between laser and nanowires.

MOTIVATION

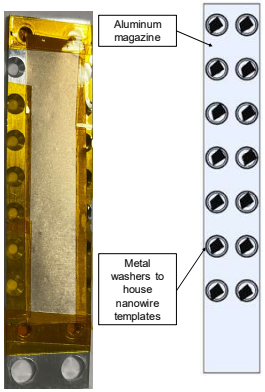
- Measuring the absorption of 10^{20} W/cm² pulse
- Comparing the light absorption of nanowire targets against flat foil targets of the same base material
- Using an Integrating Sphere to measure scattering of incident light.



EXPERIMENT DETAILS

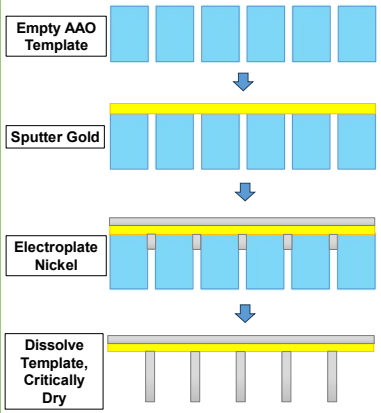
- Nanowires are irradiated with a relativistic femtosecond laser pulse, and light absorption values are compared to flat targets
- The process of shooting nanowires and flat targets is shown below

Fabricate 13mm arrays of nanowires → Break the nanowire samples into multiple pieces → Mount nanowire targets on washers → Image wires under optical microscope → Mount washers on aluminum magazine housing → Fixture magazine into the Integrating Sphere

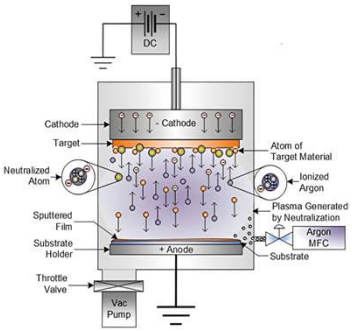


- Cut-out samples are placed on washers due to the central hole/opening, which ensures effective transmittance of photons on the back reflection detectors, subsequently ensuring consistent, and reliable results.
- Imaging the wires under an optical microscope serves as a preliminary validation step to assess whether the wires are uniformly spaced or bunched together.
- Target holders need to be a certain distance away from each other ensuring the heat from irradiation doesn't affect adjacent samples.

NICKEL NANOWIRES



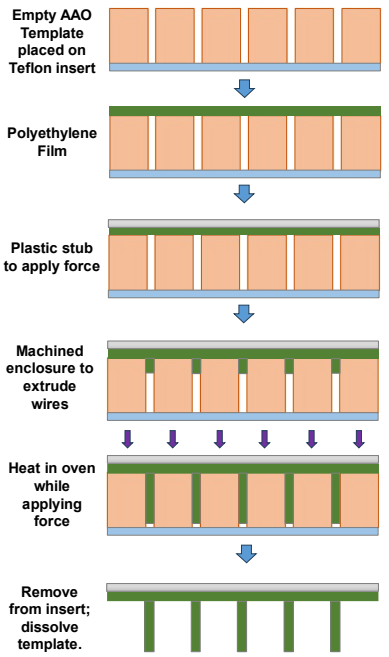


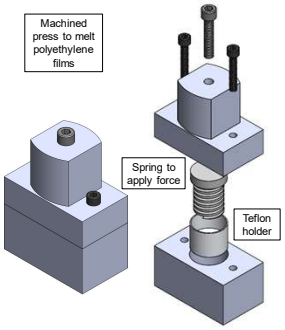


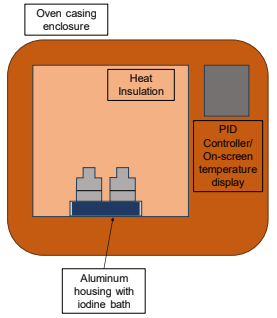
Parameters

- Nanowire characteristics depend on several parameters.
- Diameter, pore density, and length affect laser interaction data.
- Wires are dried in a critical point dryer using CO_2.
- Higher pore density allows more wires per micron of template.
- High-density wires require constant pipetting.

POLYETHYLENE NANOWIRES





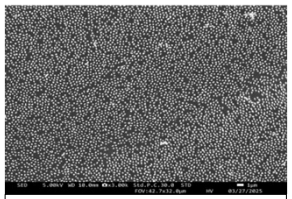


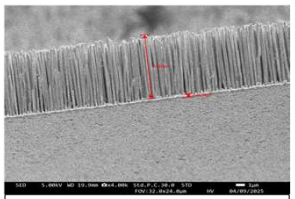
Parameters

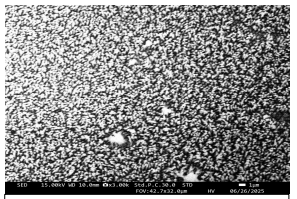
- CH2 wires are extruded through a lattice in an AAO Template
- Extrusion occurs in a 200°C oven
- Wire lengths are limited to 5µm or 10 µm.
- Cleaning is similar to nickel nanowires but excludes nitric acid.
- A thermocouple monitors the press until it reaches 140°C.
- Wires are dissolved in NaOH at 50°C.

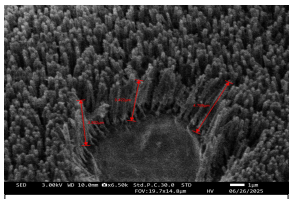


RESULTS OF NANOWIRE FABRICATION


Top View


Side View


Top View


Side View



ACKNOWLEDGEMENT

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J.J. Rocca, M.G. Capeluto, R.C. Hollinger, S. Wang, Y. Wang, G.R. Kumar, A.D. Lad, A. Pukhov, and V.N. Shylyaptsev, Ultra-intense femtosecond laser interactions with aligned nanostructures, OPTICA, 11: 510542, (2024)